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CHALLENGE - 10 - CRYPTOGRAPHY

BSIT-3

rotation

Medium

Cryptography

picoCTF 2023

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Description

You will find the flag after decrypting this file
Download the encrypted flag [here](#).

Hints ?

1

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The **Rotation** challenge in picoCTF 2023 is a beginner-friendly cryptography task that involves decoding a message encrypted using a rotation cipher (also known as a Caesar cipher).

```
encrypted.txt
File Edit View H1 100% Unix (LF)
xqkwKBN{z0b1b1wv_13kzgb31_9491n111}
```

Objective

The goal is to decrypt the provided encoded file and retrieve the flag in the format:
`picoCTF{r0tat1on_d3crypt3d_949af1a1}`

Approach

Upon downloading the encrypted file, the contents appear as unreadable text. This suggests the use of a simple substitution cipher, specifically a rotation cipher where each letter is shifted by a fixed number in the alphabet.

Since the rotation value is unknown, the most effective method is to try all possible shifts (ROT1 to ROT25). This is considered a brute-force approach in basic cryptography.



dCode is preparing a new interface. Come test and give your feedback on the new page: [ROT Cipher!](#)

ROT CIPHER

Cryptography › Substitution Cipher › ROT Cipher

ROT CIPHER DECODER

★ ROTED TEXT
xqkwKBN{z0b1b1wv_13kzgx3l_949in11}

AUTOMATIC DECRYPTION (BRUTE-FORCE)

★ (EXPECTED) PLAINTEXT LANGUAGE: English

▶ DECRYPT

CUSTOM DECRYPTION

★ ROTATION TO USE ROT-N, N= 13

★ ALPHABET TO USE

☒ ABCDEFGHIJKLMNOPQRSTUVWXYZ (Rot13 / CAESAR)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ AND 0123456789 (Rot13+5)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789 (Rot18)

☐ 0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ (Base36)

☐ 94 PRINTABLE ASCII CHARACTERS FROM !33 TO ~126 (Rot47)

☐ FULL ASCII TABLE (128 CHARACTERS)

☐ CUSTOM ALPHABET (CASE INSENSITIVE)

1234567890QWERTYUIOPASDFGHJKLZXCVBNM

☐ CUSTOM ALPHABET (CASE SENSITIVE)

qwertyuiopasdfghjklzxcvbnm1234567890QWERTYUIOPASD...

▶ DECRYPT

See also: [Caesar Cipher](#) – [Shift Cipher](#) – [ROT-13 Cipher](#) – [ROT-47 Cipher](#)

ROT-N ENCODER

★ PLAIN TEXT TO ROTATE
dcode ROT

★ ROTATION TO USE ROT-N, N= 13

★ ALPHABET TO USE

☒ ABCDEFGHIJKLMNOPQRSTUVWXYZ (Rot13 / CAESAR)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ AND 0123456789 (Rot13+5)

☐ ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789 (Rot18)

☐ 0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZ (Base36)

☐ 94 PRINTABLE ASCII CHARACTERS FROM !33 TO ~126 (Rot47)

Results

↑↓	↑↓
[A-Z]+8	picoCTF{r0tat1on_d3crypt3d_949af1a1}
[A-Z0-9]+25	81v7VMY{abmtmc76_wevar8mew_kfktyctc}
[A-Z][0-9]+8	picoCTF{r2tat3on_d5crypt5d_161af3a3}
[A-Z0-9]+27	6zt5TKW{89krka54_uct8p6kcu_idirwara}
[A-Z0-9]+8	picoC3F{rs3a3ton_dvcr8p3vd_1w1afat}
[A-Z0-9]+23	a3x9X00{cdovoe98_ygxctaogy_mhmv0eve}
[A-Z0-9]+22	b4yaYP1{depwpfa9_zhydubphz_ninw1fwf}
[A-Z][0-9]+19	exrdRIU{g1ipi2dc_s4rgnei4s_050pu2p2}
[A-Z]+19	exrdRIU{g0ipi1dc_s3rgnei3s_949pu1p1}
[A-Z0-9]+24	92w8WNZ{bcnund87_xfwbs9nfx_lgluzdud}
[A-Z]+10	ngamARD{p0ryr1m_l_b3apwnr3b_949yd1y1}
ASCII[!-~]+5	s1frF=Ivu+ d],rqG.fubs].gZ4/4di,d,x
ASCII[!-~]+40	PICO#x&SRF:A:g0N7DiCR?P:iD7ojoAFgAgU
[A-Z0-9]+29	4xr3RIU{67ipi832_sar6n4ias_gbgpu8p8}
ASCII[!-~]+8	picoC:Fsr(ZaZ)onWd+cr_pZ+dW1,1af)a)u
[A-Z0-9]+10	ngamA1D{pq181rm_l_btap6n1tb_zuz8dr8r}

Summary

- ★ ROT Cipher Decoder
- ★ ROT-n Encoder
- ★ What is Rot cipher? (Definition)
- ★ How to encrypt using Rot cipher?
- ★ How to decrypt with Rot cipher?
- ★ What are rot variants?
- ★ Why is the ROT cipher used?
- ★ What is the Rot cipher algorithm?

Similar pages

- ★ Caesar Cipher
- ★ ROT-13 Cipher
- ★ ROT-47 Cipher
- ★ Shift Cipher
- ★ ROT1 Cipher
- ★ ROT-5 Cipher
- ★ ASCII Shift Cipher
- ★ DCode's TOOLS LIST

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Forum/Help

DISCORD

Keywords

rot, rotation, caesar, code, shift, rot13, rot47, alphabet, circular

Links

- ★ [Contact](#)
- ★ [About dCode](#)
- ★ [dCode App](#)

Solution Process

1. Open the encrypted file using a text editor.
2. Copy the ciphertext.
3. Use a rotation/caesar cipher tool (or script) to test all 25 possible shifts.
4. Scan the outputs for readable English text.
5. Identify the correct shift when the message becomes meaningful.
6. Locate the flag within the decoded text.

Key Insight

One of the rotations will clearly reveal a readable sentence containing the flag. The correct rotation stands out because it produces coherent words and follows the expected flag format.

Conclusion

This challenge demonstrates how weak simple encryption methods like rotation ciphers are, especially when brute-force techniques are applied. It also reinforces foundational skills in cryptography, such as pattern recognition and systematic testing.